



Midpoint

The options bill **nobody** shows you —
and an agent that trades under the
discipline it implies.

Alpaca AI Trading Agents Hackathon

account **PA32CGA2U1DY**

Purushottam Sahu

THE PROBLEM

Two option trades. One costs \$1. The other costs \$306.

SPY 751 P

\$1

bid 0.14 · ask 0.15

SPY 781 P

\$306

bid 13.95 · ask 17.01

Both real, both measured on live markets, both the same underlying in the same week. The bars are drawn to the same scale. **Nothing on a broker screen tells you which one you just clicked.**

One word in US securities law

“...any NMS security **other than an option.**”

17 CFR 242.600 – the definition of an NMS stock

Trade **shares**, and SEC Rule 605 forces your broker to publish an execution-quality receipt.

Trade **options**, and that rule does not apply. So **no options execution data exists** for anyone to compare against.

HOW MUCH THIS IS WORTH

Retail traders lose 3× more to the toll than to being wrong

\$6.4bn

lost to trading costs

\$2.1bn

lost to bad trades

3×

the toll, against the losses everyone talks about

Bryzgalova, Pavlova & Sikorskaya · **Journal of Finance**, 2023 · aggregate US retail options, Nov 2019 – Jun 2021

The receipt, and an agent that earns a good one

HALF ONE — MEASURE

A tool that prices any option trade from two numbers off your own broker screen.

- 65 contracts measured by paying the cost
- 2,097 quote observations recorded
- Runs in the browser — nothing is sent anywhere

HALF TWO — TRADE

An autonomous agent on a live Alpaca paper account, under 14 risk gates.

- 651 proposals judged, 623 refused
- Defined risk on every structure, always flat by 15:15
- No model. Measurement plus rules

One loop every 70 seconds. Every stage can only say no.

01

Look

VIX term structure and the SPY option chain. Permission comes from the slope, not the level.

02

Build

One candidate: short strike near 0.20 delta, loss capped by a long leg one strike out.

03

Judge

Fourteen risk gates in fixed order. Any single one can veto the trade outright.

04

Act

Send only if all fourteen agree. Close at 50% of credit. Flat by 15:15 ET.

The agent claims **no forecasting edge**. Its intelligence is in what it declines — and it declines **96%** of what it considers.

Fourteen gates, mapped onto SEC Rule 15c3-5

G0-account	wrong account	G9-liquidity	quotes wider than \$0.20
G1-kill-switch	the HALT file exists	G10-duplicate	same structure inside 60s
G2-defined-risk	loss not capped by a long leg	G11-throttle	over 20 orders a minute
G13-escalation	25 malfunction-class rejects	G3-trade-size	risk over 0.25% of equity
G4-daily-loss	session down 1%	G6-aggregate	open risk over 1.5%
G5-drawdown	2.5% below equity peak	G7-fat-finger	over 10 contracts
G12-clock	ODTE after 14:00 ET	G8-price-collar	limit 10% off fair value

Pure kernel – no network, no clock. **23 unit tests + 12 invariants × 10,000 random states.** G13 encodes Knight Capital: 97 warnings, nobody acted, \$460m in 45 minutes.

Every system measures the trades it made. This one measures the trades it refused.

138

refusals priced by settling each one at
expiry

49

would have **lost money** — refusing
saved \$3,172

+\$2,491

net value of **saying no**

Conventional transaction-cost analysis structurally cannot do this: it is built from executed orders and is **blind to every trade never made**. Storing the full market snapshot behind each refusal is what closes that gap.

\$13 on \$100,000. Here is why that is the number.

Over five sessions, a skilled strategy and a lucky one produce numbers you cannot tell apart.

Grinold & Kahn: SE of an information ratio is $1/\sqrt{\text{years}}$ — **~16 years** to establish top-quartile skill

So a big five-day P&L is not evidence of skill. It is evidence of **variance**.

- 5 trades across 2 sessions
- Every one closed at the agent's own target
- No losing round trip
- Every dollar attributed by the **order that caused it**
- Research spending split out, never blended
- A **\$0.30** gap traced to 12 legs at \$0.025 — and recorded as a residual

A number is only worth something if the method would also have reported a negative

WE DO NOT CLAIM

- That this beats the market
- That five sessions of P&L means anything
- That our regime signal is Johnson's PC2 — it is a two-point proxy
- That paper fills would survive real market impact

PUBLISHED ANYWAY

- **Two null results** — the microprice does not beat the plain mid here ($p = 0.30$)
- **One withdrawn result**, retracted after finding our own error
- A venue audit proving paper P&L is earned where liquidity is free
- Every figure recomputable offline from published data



**Options traders pay \$6.4bn a
year in costs like these.
Nobody sends them the bill.**

This does.

app `sahuongithub.github.io/midpoint`

code `github.com/sahuongithub/midpoint`

account `PA32CGA2U1DY`